

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

EE 19-209-0402 CIRCUITS, SIGNALS AND SYSTEMS

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Classify basic signals – impulse, step, ramp, exponential and sinusoidal signals, operations of signal and properties of continuous time systems.

CO2: Analyse one sided and two sided frequency spectrum of periodic signals using Fourier method of waveform analysis.

CO3: Design passive filters.

CO4: Synthesize L-C and R-C circuits using Foster and Cauer Forms.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer **ALL** questions)

(8 × 3 = 24) Marks BL CO PO

- | | | | | |
|---|---|----|---|---|
| I. (a) Distinguish between causal and non-causal systems. | 3 | L2 | 1 | 1 |
| (b) Check the linearity of the following system | 3 | L5 | 1 | 1 |
| $\frac{dy}{dt} + t^2 y(t) = 2x(t)$ | | | | |
| (c) Explain Dirichlet conditions for the existence of a Fourier series, focusing on how these conditions ensure the representation of a periodic function. | 3 | L1 | 2 | 1 |
| (d) Find the inverse Fourier transform of frequency shifted impulse signal, $\delta(\Omega - \Omega_0)$ if $\delta(\Omega)$ has its inverse fourier transform as $\frac{1}{2\pi}$. | 3 | L3 | 2 | 1 |
| (e) What are the important parameters that characterize a typical Filter? | 3 | L2 | 3 | 1 |
| (f) Derive the frequency of infinite attenuation in an m-derived high pass filter. | 3 | L3 | 3 | 1 |
| (g) Explain what a hurwitz polynomial is and describe its key properties. | 3 | L2 | 4 | 1 |
| (h) Draw the pole zero diagram for the given network function I(S) and hence obtain i(t) | 3 | L2 | 4 | 1 |

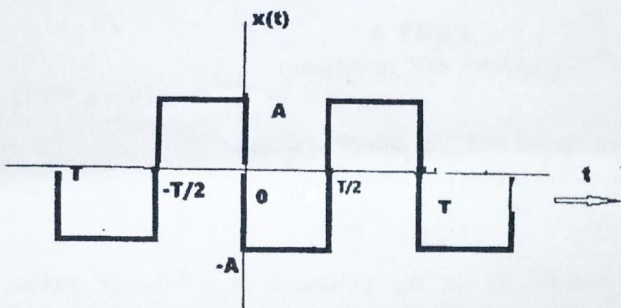
$$I(S) = \frac{20S}{(S+5)(S+2)}$$

(P.T.O.)

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PART B

(4 × 12 = 48)

		Marks	BL	CO	PO
II.	Classify the following signals into appropriate categories based on the given criteria: continuous-time/discrete-time, deterministic/non-deterministic, periodic/apperiodic and even/odd. Provide reasoning for each classification.	12	L3	1	1
	(i) $x_1(t) = e^{-t}$ for $t \geq 0$				
	(ii) $x_2(n) = \sin(\Pi n)$				
	(iii) $x_3(t) = u(t) - u(t-1)$				
	OR				
III. (a)	Sketch the following signal. $u(-2t+2)$	6	L3	1	1
(b)	Determine the output response $y(n)$ if $x(n) = \{1, 2, 3, 2\}$ $h(n) = \{1, 2, 2\}$	6	L3	1	1
IV. (a)	Find the trigonometric fourier series of the given continuous time square wave $x(t)$.	6	L2	2	1
					
(b)	The input and output of a causal LTI system is related by the differential equation. $\frac{d^2}{dt^2} y(t) + 6 \frac{d}{dt} y(t) + 8y(t) = 2x(t)$ Determine impulse response of the system.	6	L3	2	1
	OR				
V. (a)	Using the standard transforms and properties find Fourier transforms of the following signals.	6	L3	2	1
	(i) $x(t) = te^{-2t} u(t)$.				
	(ii) $x(t) = \sin(2\Pi t) e^{-t} u(t)$.				
(b)	Determine the signal which corresponds to the following Fourier transform. $x(e^{j\omega}) = (e^{-j\omega})[0.5 + 0.5 \cos \omega]$	6	L3	2	1

(Continued)

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		Marks	BL	CO	PO
VI.	(a) Obtain the variation of the following with respect to frequency for a constant k low pass filter. (i) characteristic impedance (ii) attenuation constant.	5	L2	3	1
	(b) Design an m-derived T section low pass filter having cut off frequency of 7 kHz, design resistance, $R_o = 600 \Omega$ and frequency of infinite attenuation, $f_\infty = 10.5$ kHz.	7	L3	3	1
OR					
VII.	(a) Design a prototype filter π section to match with a line having characteristic impedance of 500Ω and pass frequencies upto 5 kHz.	6	L5	3	1
	(b) List the limitations of a prototype filter.	3	L3	3	1
	(c) Draw the block diagram of a composite filter which can overcome the above limitations.	3	L5	3	1
VIII.	(a) Check positive realness of the following function. $Z(S) = \frac{S^3 + 5S^2 + 9S + 3}{S^3 + 4S^2 + 7S + 9}$	6	L5	4	1
	(b) Find out the second order Cauer network of the following function. $Z(S) = \frac{2S^2 + 8S + 6}{S^2 + 8S + 12}$	6	L6	4	1
OR					
IX.	Find the first foster form of network for the following impedance function. $Z(S) = \frac{S(S^2 + 2)}{(S^2 + 1)(S^2 + 3)}$	12	L6	4	1

Bloom's Taxonomy Levels

L1 – 2.5%, L2 – 19.17%, L3 – 48.33%, L5 – 15%, L6 – 15%.
